

Jinyoung Park

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SUMMARY

Ph.D. Candidate in Biomedical Engineer with an MS in Electrical & Computer Engineering, specializing in biodegradable implantable electronics and microelectrode arrays for optogenetic recording. Expert in cleanroom microfabrication, piezoelectric pressure sensing systems, and wireless Bluetooth integration for real-time physiological monitoring.

SELECTED RESEARCH EXPERIENCES

Research Specialist & Graduate Research Assistant

April 2021 – Present

University of Connecticut, Department of Biomedical Engineering, Storrs, CT

- Developed implantable piezoelectric pressure sensor system for post-surgical knee joint loading monitoring
- Research on brain cancer treatment with biodegradable ultrasound transducer and targeted gene therapy

Research Assistant

September 2018 – August 2020

University of Massachusetts, Department of Electrical and Computer Engineering, Amherst, MA

- Engineered PEDOT:PSS-coated graphene microelectrode arrays to achieve high-fidelity optogenetic recording and enhanced charge injection for chronic neurostimulation

Graduate Research Assistant

March 2016 – February 2018

University of Seoul, Department of Electrical and Computer Engineering, Seoul, South Korea

- Developed design methodologies for sub-10-nm stacked nanowire FETs, implementing structural optimization to suppress process-induced random variations and enhance device-level reliability
- Project supervisor for undergraduate research program: Responsible for entire project. Supervised undergraduate students. Work involved in the experimental research: piezoelectric-oxide-semiconductor field-effect transistor using negative capacitance effect.

TECHNICAL SKILLS

- Materials & Bioelectronics: Polylactic Acid, Biodegradable Polymers, Graphene, PEDOT:PSS
- Fabrication & Characterization: Electrospinning, Electrochemical Impedance Spectroscopy (EIS), Cyclic Voltammetry (CV), Microfabrication (cleanroom), SEM, AFM, Signal Processing
- Software: Sentaurus TCAD simulation, MATLAB, C/C++

EDUCATION

University of Connecticut (UConn), Storrs, CT

September. 2026 (anticipated)

Doctor of Philosophy (Ph.D.) in Biomedical Engineering

University of Seoul, Seoul, Korea

February 2018

Master of Science (M.Sc.) in Electrical and Computer Engineering

University of Seoul, Seoul, Korea

February 2016

Bachelor of Science (B.Sc.) in Electrical and Computer Engineering

AWARDS

- Student Travel Achievement Recognition (STAR) Award, *Society of Biomaterials* *March 2026*
- Conference Participation Award, *University of Connecticut* *July 2024*
- Schriever Education Award, *Society of Plastics Engineers* *June 2024*

SELECTED SCHOLARLY PUBLICATIONS AND PRESENTATIONS

Publications (4 out of 15)

- “Wireless Knee Joint Monitoring Utilizing Biodegradable Single-ended Pressure Sensor in Osteoarthritis Management” *Science Advances*, (ACCEPTED; May, 20206)
- “Electrically active biomaterials for stimulation and regeneration in tissue engineering,” *Journal of Biomedical Materials Research Part A*, 113:e37871, Jan. 2025.
- "Highly piezoelectric, biodegradable, and flexible amino acid nanofibers for medical applications," *Science Advances*, vol. 9, no. 24, Jun. 2023.
- “Low-Impedance Low-Artifact PEDOT:PSS-Coated Graphene Microelectrodes Towards High Density Optogenetic Electrophysiology,” *IEEE Electron Device Letters*, vol. 41, no. 8, Aug. 2020.

Presentations (3 out of 11)

- “Advancing Brain Cancer Treatment with Biodegradable Ultrasound Transducer and Targeted Gene Therapy,” *Society for Biomaterials (SFB) 2026 Annual Meeting & Exposition*, 2026 (submitted).
- “Extended-Lifetime Biodegradable Pressure Sensor for Knee Joint Load Monitoring in Osteoarthritis Treatment,” *Biomedical Engineering Society (BMES) 2025 Annual Meeting*, 2025.
- “High-resolution neurostimulation and optogenetic electrophysiology with PEDOT:PSS-coated graphene,” *The 66th International Electron Devices Meeting Conference (IEDM)*, 2020.